NetraJyoti AI

Project Delivery, Ethical Considerations and Lean Canvas

This supporting capstone document explains how NetraJyoti AI can be delivered responsibly: why the product is viable, how the team will work, how work will be tracked in Jira, and how safety, privacy, and inclusion are protected.

Product boundary. NetraJyoti gives Bengali-first eye-care guidance and service navigation. It helps users understand safe next steps; it does not diagnose disease or replace a clinician.

Document item	Value
Product	NetraJyoti AI MVP
Recommended delivery approach	Lean UX + Agile Kanban
Planning horizon	12-week capstone sample plan
Work tracking	Jira project NET; Epic NET-1 and associated stories
Primary users	Rural Bengali-speaking eye-care seekers, caregivers, ASHA/community-health workers, outreach coordinators
Ethical priority	Safety-first, non-diagnostic, privacy-preserving care navigation

1. How the deliverables fit together

The five requested items are one delivery story, not separate exercises. The Lean Canvas explains the product opportunity; ethical considerations define what the product may and may not do; the methodology explains how uncertainty will be reduced; the project plan turns that approach into milestones and ownership; and the Jira board provides visible execution tracking.

Deliverable	Question answered	Output
Lean Canvas	Why does this product matter and how could it create value?	Nine-block product and viability model.
Ethical considerations	What safety, privacy, inclusion, and accountability limits apply?	Safeguards and governance checks.
Methodology	How will the team learn and deliver safely?	Lean UX discovery plus Kanban delivery flow.

Deliverable	Question answered	Output
Sample project plan	When will each activity happen and who owns it?	Sequenced 12-week plan with milestones.
Jira board	How is work made visible and managed?	Backlog/To Do/In Progress/Testing/Done workflow and evidence screenshot.

2. Lean Canvas

This uses the default Lean Canvas structure. Assumptions should be tested through real user interviews, usability sessions, clinical/content review, and a small monitored pilot.

Lean Canvas block	NetraJyoti application	Assumption to validate
Problem	People in rural West Bengal may delay eye care because they do not know whether symptoms are serious, which information to trust, or where to go. Cost, travel, waiting time, low digital confidence, and language barriers add friction.	Existing alternatives: family advice, informal care, forwarded messages, generic web search, and direct travel without service certainty.
Customer segments	Primary: Bengali-speaking rural users with an eye concern. Secondary: caregivers who help make decisions. Tertiary: ASHA/community-health workers and outreach coordinators.	Early adopters: users/caregivers who have a phone, a current eye concern, and uncertainty about the right care route.
Unique value proposition	“Bengali-first safe next-step guidance for eye problems.” NetraJyoti reduces confusion before treatment by turning fixed symptom inputs into clear, non-diagnostic care navigation.	High-level concept: a safety-first Bengali voice/text guide to the appropriate eye-care service.
Solution	Fixed symptom selection; reviewed urgent-warning check; deterministic urgent/routine/human-support routing; district/town service direction; family sharing; optional controlled AI caregiver summary.	MVP deliberately excludes diagnosis and treatment advice.
Channels	Eye camps, vision centres, ASHA/community-health workers, NGO/eye-care partners, posters/QR codes, local WhatsApp sharing through caregivers, and pilot referrals.	Start with partner-supported community outreach, then assess digital referral channels.
Revenue streams / funding	For a social-impact pilot: NGO/CSR sponsorship, eye-care partner programme funding, public-health grant support, or service-provider implementation agreements.	No user-facing payment is assumed in the MVP.
Cost structure	Product design/research, clinical and Bengali-content review, frontend/backend development, cloud/database hosting, AI usage when enabled, service-directory verification, usability testing, and outreach operations.	The controlled-AI feature must remain optional to contain both cost and safety risk.
Key metrics	Journey completion; route distribution; time to reach recommended action; service-direction use; caregiver share/copy rate; comprehension/usability score; AI fallback rate; verified-service data freshness.	Never use clinical outcome claims without a validated study design.
Unfair advantage	Trusted local eye-care and community-health partnerships, reviewed Bengali content, curated service data, and a safety-first workflow tailored to the rural West Bengal context.	The advantage must be maintained through regular service-data verification and user research.

3. Recommended methodology: Lean UX + Agile Kanban

Use a combined approach: Lean UX for problem discovery and evidence gathering, followed by Agile Kanban for visible, continuously prioritised delivery. This is appropriate because NetraJyoti has both product uncertainty (language, comprehension, service-navigation behaviour) and safety constraints that require short, reviewable increments.

Practice	How it is applied to NetraJyoti	Why it fits
Lean UX discovery	Interview users, caregivers, ASHA workers, and outreach coordinators; map current care-seeking journey; test wireframes in Bengali; define hypotheses before building.	Prevents building a text-heavy or clinically unsafe workflow without evidence.
Kanban delivery	A prioritised backlog and visual workflow (To Do, In Progress, Testing / Content Review, Done), with regular review, improvement actions, and release-ready increments.	Makes work visible, limits work in progress, and enables continuous prioritisation and safe release checks.
Safety/content review	Review symptom labels, urgent criteria, Bengali wording, service-data disclaimers, and AI fallback before release.	Health content requires defined approval gates rather than only engineering sign-off.
Usability testing	Run short task-based sessions on Android-sized devices, slow networks, and voice/location denied states.	Validates accessibility and practical completion, not just feature delivery.
Pilot learning loop	Measure outcomes, collect feedback, fix high-risk friction, and revalidate content/service data before expansion.	Supports evidence-led iteration and avoids overclaiming MVP impact.

3.1 Example two-week sprint cadence

Day	Activity	Evidence / output
1	Planning and safety/content review	Prioritised Jira stories, acceptance criteria, and risk check.
2–8	Design, build, and test	Working increment; unit/component tests; reviewed copy.
9	Internal demo and QA	Defect list, route regression result, service-data confirmation.
10	Flow review and improvement planning	Stakeholder feedback, updated backlog, and improvement actions.

4. Sample 12-week project plan

This is a sample capstone plan. The dates can be replaced with your actual submission calendar. Each phase has a clear decision or artefact so that progress is measurable.

Week	Focus	Key activities	Owner	Milestone
1	Discovery and evidence	Interview/research plan; current journey; problem validation; safety assumptions.	Product owner + researcher	Problem evidence and risk log approved.
2	Personas and Lean Canvas	Personas, jobs-to-be-done, Lean Canvas, success metrics.	Product owner	Target users and value proposition agreed.
3	User journey and information architecture	End-to-end branches, content inventory, wireframe requirements.	Product + UX	Urgent/routine/ human flow reviewed.
4	Wireframes and usability prototype	Bengali-first prototype; test script; user feedback.	UX + product	Usability findings prioritised.
5	Technical design and backlog	HLD, Technical PRD, API/data design, Jira stories and estimates.	Tech lead + product	MVP backlog ready.
6	Core intake and routing	Consent, symptoms, urgent-sign questions, deterministic route API.	Frontend + backend	Core safe-route tests pass.
7	Urgent and routine navigation	5A route, 5B–7 routine route, service-direction demo data.	Frontend + backend	No-dead-end journey demo.
8	Support, sharing, and accessibility	5C route, family sharing, patient-detail consent, responsive UX.	Frontend + QA	Share/fallback/ device tests pass.
9	Controlled AI and resilience	Feature flag, backend-only OpenAI integration, validator and fallback.	Backend + safety reviewer	AI-off/error fallback confirmed.
10	Pilot readiness and content governance	Service-data verification, privacy review, analytics plan, release checklist.	Outreach + product + QA	Pilot gate decision.
11	Pilot/usability validation	Task-based sessions; collect completion and comprehension evidence.	Research + QA	Findings and prioritised fixes.
12	Final refinement and capstone pack	Fix priority issues; final PRD/HLD/Jira screenshot/demo/report.	Whole team	Submission-ready package.

Dependency note. Work on live service data, public pilot, and expanded AI capability must not begin until clinical/content review, privacy review, and operational ownership are confirmed.

5. Jira Kanban board, Epic and dashboard evidence

Use Jira project NET and the NetraJyoti epic/story structure to demonstrate Agile Kanban delivery. The board should reflect actual delivery state, not merely a list of issues, using its configured workflow columns.

Board element	Recommended configuration
Board name	NetraJyoti AI MVP Delivery Board
Source	NET project; filter for the NetraJyoti epic and its child stories.
Columns	Backlog → To Do → In Progress → Testing / Content Review → Done.
Swimlanes	Optional: urgent care, routine care, human support, platform/content.
Card fields	Issue key, summary, assignee, priority, story points, sprint, and epic link.
Useful quick filters	Urgent path; routine path; human support; high priority; blocked; content/service-data updates.
Dashboard evidence	Add a Jira listing or Filter Results gadget for the NetraJyoti epic, plus sprint progress and priority view where available.

5.1 Jira delivery links

Jira Epic: NET-1 - NetraJyoti Epic MVP

Jira dashboard: **NetraJyoti dashboard**

6. Ethical considerations and safeguards

NetraJyoti operates in a health-adjacent setting where unsafe communication can cause delay or false confidence. Ethical design must therefore be visible in requirements, user interface, content review, technical architecture, and operations.

Ethical consideration	Risk	Required safeguard
Non-diagnosis and informed expectations	The app may help users decide a safe next step but must not claim to identify a disease, prescribe treatment, or replace a clinician.	Use clear value-led consent; keep disease labels and medication advice out of content and AI output.
Urgency and harm prevention	Under-triage could delay care; over-triage can create fear and burden services.	Use clinician-reviewed deterministic rules, version control, tests for warning-sign combinations, and conservative urgent wording.
AI safety and accountability	Generative output can invent clinical advice, service details, or urgency.	AI is optional, backend-only, routine-route-only, feature-flagged, schema/policy validated, and always has approved fallback.
Privacy and consent	Names, addresses, phone numbers, voice, and location are sensitive.	Collect only optional message fields; require explicit share consent; keep data client-side in MVP; do not log/store PII by default.
Location consent	Precise location can reveal sensitive context and must never be assumed.	Ask only after a clear user action; provide district/town alternative; do not retain coordinates.
Language, literacy, and accessibility	English-only or text-heavy interaction can exclude the intended community.	Bengali-first labels, smaller English support text, large targets, familiar wording, optional voice, and plain-language testing.
Equity and bias	The product may work differently across age, gender, disability, literacy, device access, and districts.	Test with diverse rural users/caregivers; offer human support; avoid making AI or smartphone access a prerequisite for guidance.
Service-data accuracy	Outdated clinic/camp/phone information may cause unnecessary travel or delay.	Show verified-at/demo status; assign outreach ownership; define update cadence and escalation for stale entries.
Human escalation and autonomy	Users should not be forced to rely solely on automation.	Provide ASHA, PHC, clinic, family-sharing, restart, and service-navigation options in every appropriate branch.
Transparency and feedback	Users and stakeholders need to understand limitations and report problems.	Explain purpose and privacy in simple language; offer feedback/help path; maintain non-identifying audit events and review process.

6.1 Governance and release gate

- Clinical/content reviewer approves rules, urgent copy, Bengali scripts, and changes affecting safety.
- Outreach coordinator verifies service/camp details and owns freshness updates.
- Product owner approves scope and evidence claims; engineering owner controls feature flags and deployment.
- Privacy/security review confirms no PII/secrets in logs, browser bundle, analytics, or Git.
- Pilot release requires tested fallback for AI, location, voice, and service lookup failures.

7. Submission checklist

Item	Submission-ready evidence
Lean Canvas	Completed Section 2, validated assumptions clearly marked.
Methodology	Section 3 explains Lean UX + Agile Kanban and sprint cadence.
Project plan	Section 4 with actual dates/owners substituted if required.
Jira board	Real screenshot inserted in Section 5.1 after board capture.
Ethics	Section 6 safeguards linked to the PRD/HLD and live product behavior.
Supporting artefacts	PRD, HLD, wireframes, Jira board/dashboard, and GitHub repository links included in final pack.